Canonical Research Designs I: Difference-in-Differences: Staggered Timing and Event Studies

Yagan (2015)

- Yagan (2015) tests whether the 2003 dividend tax cut stimulated corporate investment and increased labor earnings

- Big empirical question for corporate finance and public finance

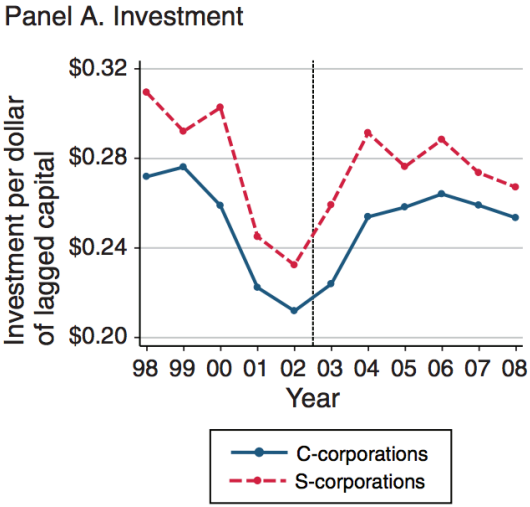
- No direct evidence on the real effects of dividend tax cut
  - real corporate outcomes are too cyclical to distinguish tax effects from business cycle effects, and economy boomed

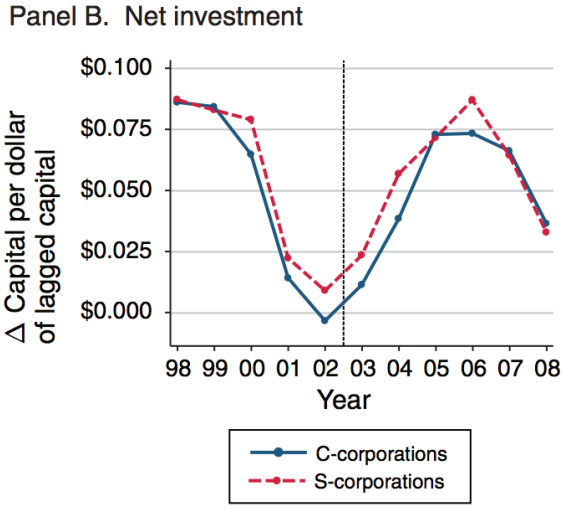
- Paper uses distinction between "C" corp and "S" corp designation to estimate effect
  - Key feature of law: S-corps didn't have dividend taxation

- Identifying assumption (from paper):

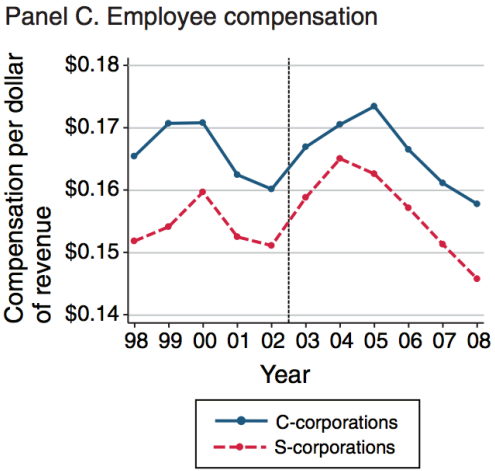
  The identifying assumption underlying this research design is not random assignment of C- versus S-status; it is that C- and S-corporation outcomes would have trended similarly in the absence of the tax cut.

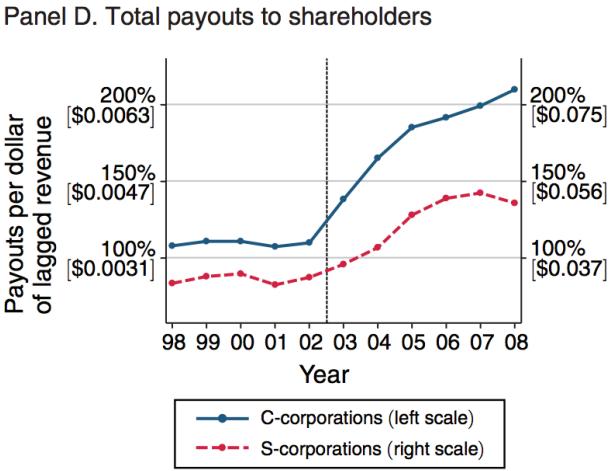
Investment Effects (none)





Employee + Shareholder effects (big)





Key Takeaway + threats

- Tax reform had zero impact on differential investment and employee compensation
- Challenges orthodoxy on estimates of cost-of-capital elasticity of investment
- What are underlying challenges to identification?
  1. Have to assume (and try to prove) that the only differential effect to S- vs C-corporations was through dividend tax changes
  2. During 2003, could other shocks differentially impact?
     - Yes, accelerated depreciation – but Yagan shows it impacts them similarly.
- Key point: you have to make *more* assumptions to assume that zero **differential** effect on investment implies zero **aggregate** effect.

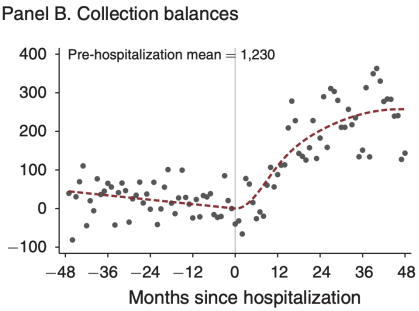
Event study

- Two important cases with these staggered timing dind (event studies)
  - There exists a never-treated group who is a potential control group
  - Everyone is treated eventually (No group is a "pure control")

- The older approach to estimate this model was:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{s=L_0, s\neq -1}^{L_1} 1(t - T_i = s)\mu_s$$

- Without a true control group, can't have both time fe, unit fe, and the full set of relative effects
- Need to exclude both the baseline period AND at least some periods outside the treatment window



- Dobkin et al. (2018)
- Comparison is between those not yet hospitalized and those hospitalized

Event study continued

- The necessary assumptions are the same (or similar) what we discussed last class
- Parallel trends

$$E(Y_{i,t}(\infty) - Y_{i,t'}(\infty)|G_i = g) = E(Y_{i,t}(\infty) - Y_{i,t'}(\infty)|G_i = g), \forall g, g', \text{and} t, t' \quad (1)$$

- Turns out, all of the groups need to be parallel.
- That might be a bad assumption (e.g. very far apart from one another)
  - Can be weakened in some cases, but only partially
- No anticipation:

$$Y_{it}(g) = Y_{it}(\infty) \forall t < g \quad (2)$$

Contamination Bias in event studies

- Sun and Abraham (2021) show that if the dynamic path of treatment is the same across cohorts (g), then the coefficient from the TWFE model will correctly estimate the period ATT

$$\tau_{it}(g) = \sum_{s \geq 0} \tau_s 1(t - g = s)$$

- If not, then there is g specific heterogeneity in paths. This creates issues:
  - Violate the pre-trend test as the use of "excluded" periods potentially contaminates pre-periods
  - Mismeasure the dynamic effects
  - Additional untestable assumptions are required as we allow for more types of heterogeneity

Issues in Diff-in-Diff - Negative Weighting vs. Contamination Bias

- There are two distinct issues in staggered timings:
  1. Goodman-Bacon (2021) and others show that the aggregated TWFE estimate can put *negative* weight on some treatment cohorts, thereby giving nonsensical estimands
  2. Sun and Abraham (2021) and others show that the *dynamic* TWFE estimates can be *contaminated* across time

- See discussion in Goldsmith-Pinkham, Hull and Kolesar (2022) appendix for analogy to broader linear regression issue

- Key point: TWFE linear regression is misspecified

Bailey and Goodman-Bacon (2015)

- Paper studies impact of rollout of Community Health Centers on mortality
  - Idea is that CHCs can help lower mortality (esp. among elderly) by providing accessible preventative care

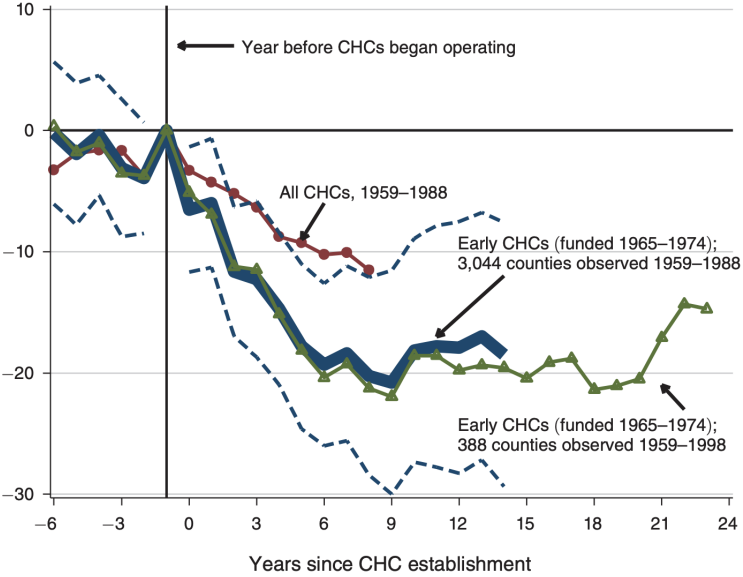
- Exploit timing of implementation of CHCs

  Our empirical strategy uses variation in when and where CHC programs were established to quantify their effects on mortality rates. The findings from two empirical tests support a key assumption of this approach—that the timing of CHC establishment is uncorrelated with other determinants of changes in mortality.

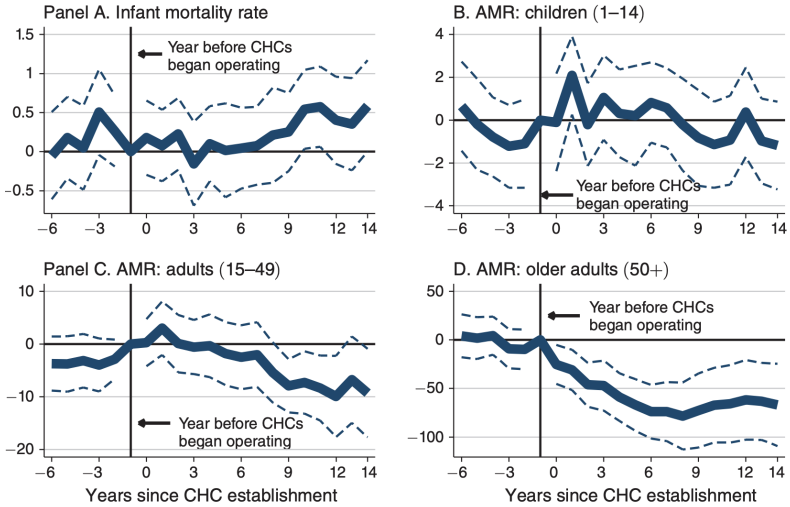
- Issue is that CHCs tend to be done in places

- Since CHCs are started in different places in different time periods, we estimate effects in *event-time*, e.g. relative to initial rollout.

Negative effect on mortality



Negative effect on mortality, particularly among elderly



Key takeaways

- Since the policy changes are staggered, we are less worried about effect driven by one confounding macro shock.
- Easier to defend story that has effects across different timings
  - Also allows us to test for heterogeneity in the time series
- Still makes the exact same identifying assumptions – parallel trends in absence of changes

But a big issue emerges when we exploit differential timing

- We have been extrapolating from the simple pre-post, treatment-control setting to broader cases
  - multiple time periods of treatment

- In fact, in some applications, the policy eventually hits everyone – we are just exploiting differential timing.

- If we run the "two-way fixed effects" model for these times of DinD

$$y_{it} = \alpha_i + \alpha_t + \beta^{DD} D_{it} + \epsilon_{it} \tag{3}$$

  what comparisons are we doing once we have lots of timings?

- Key point: is our *estimator* mapping to our *estimand*?

- Well, what's our estimand?

What is our estimand with staggered timings?

- There are a huge host of papers touching on this question

- Callaway and Sant'anna (2020) propose the following building block estimand:

$$\tau_{ATT}(g,t) = E(Y_{it}(1) - Y_{it}(0)|D_{it} = 1 \forall t \geq g), \tag{4}$$

  the ATT in period t for those units whose treatment turns on in period g.
  - In the 2x2 case, this was exactly our effect!
  - This paper assumes absorbing treatment, but can be weakened in other papers (de Chaisemartin and d'Haultfoeuille (2020) discuss this)

- It seems very reasonable that for our overall estimand, we want some weighted combinated of these ATTs

- Callaway and Sant'anna (2020) highlight two ways to identify the above estimand:
  1. Parallel trends of treatment group with a group that is "never-treated", $G_i = \infty$
  2. Parallel trends of treatment group with the group of the "not yet treated", $G_i \geq t + 1$

- Using these estimands, C&S provide a very natural set of potential ways to aggregate these estimands up

Wait what happened to TWFE?

- It turns out that the logic of the TWFE does not naturally extend to differential timings
  - Recall that from our discussion of linear regression, regression is great because it does a variance weighted approximation:

$$\tau = \frac{E(\sigma_D^2(W_i)\tau(W_i))}{E(\sigma_D^2(W_i))}, \qquad \sigma_D^2(W_i) = E((D_i - E(D_i|W_i))^2|W_i)$$

  - It turns out that in the panel setting with staggered timings, these weights are not necessarily positive

- Why? The predicted value of $D_{i,t}$, $\hat{D}_{it}$, may be greater than 1 because $E(D_i|W_i)$ is the model is incorrectly specified by the two-way fixed effects:

$$\hat{\beta}_{post} = \frac{\sum_{i,t}(D_{it} - \hat{D}_{i,t})Y_{i,t}}{\sum_{i,t}(D_{it} - \hat{D}_{i,t})^2}$$

Misspecification in conditional expectation

- An example from GPHK (2022): Three time periods, and three groups of units: treated early in period 2 (n_E), treated late in period 3 (n_L), and never (n_N)

- The weights in the estimands on the treatment effects will be

$$\lambda(j, t = 3) = \frac{n_E + 2n_N}{\kappa}, j \in L \tag{5}$$

$$\lambda(j, t = 2) = \frac{n_N + 2n_L}{\kappa}, j \in E \tag{6}$$

$$\lambda(j, t = 3) = \frac{n_N - n_L}{\kappa}, j \in E, \tag{7}$$

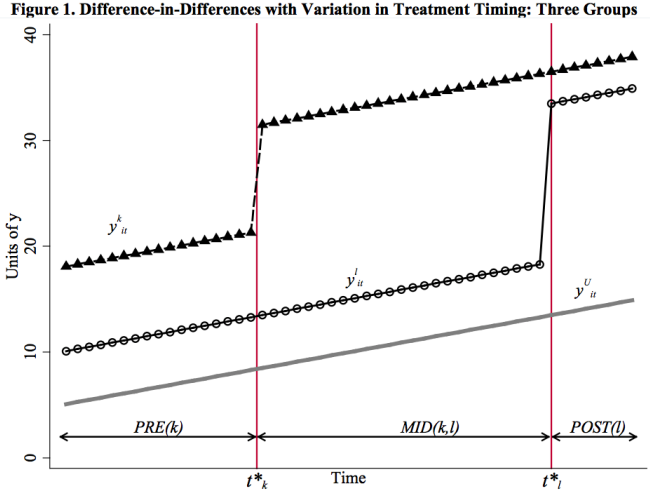
  κ just scaling. Hence, first two always get positive weight, but the ATT from period 3 of the early adopters is negative if the never-adopters is small relative to late adopters

- Key insight from several papers: with staggered timings + heterogeneous effects, the TWFE approach to DinD can put negative weight on certain groups' TE
  - Serious issue for interpretability

- This is *solveable*. Merely a construct of being overly casual with estimator definition

Goodman-Bacon 2x2 comparisons

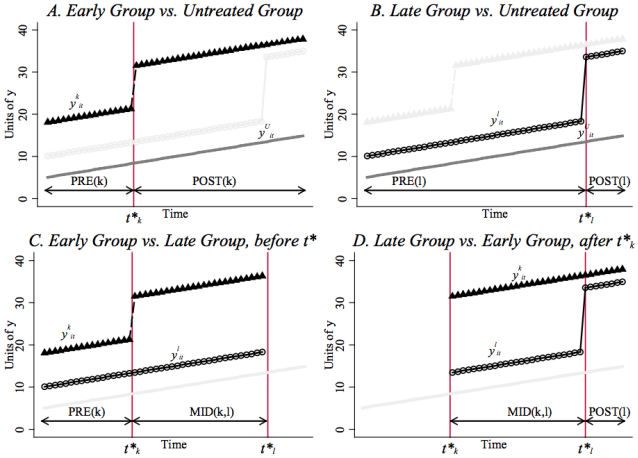
- Consider two staggered treatments and a never-treated group
- What does the TWFE estimator estimate?



Goodman-Bacon 2x2 comparisons

- Four potential comparisons that can be made
- turns out that TWFE DD estimator (pooled) is the weighted average of all 2x2 comparisons
- These weights end up putting a high degree of weight on units treated in the middle of the sample (since they have the highest variance in the treatment indicator!)

Figure 2. The Four Simple (2x2) Difference-in-Differences Estimates from the Three Group Case



Goodman-Bacon 2x2 comparisons

- The weighting becomes problematic if the effects vary over time – if the effects are instantaneous and time-invariant, the weights are all positive

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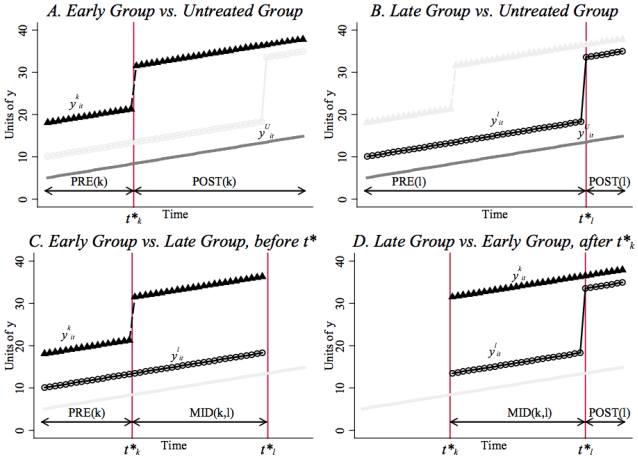


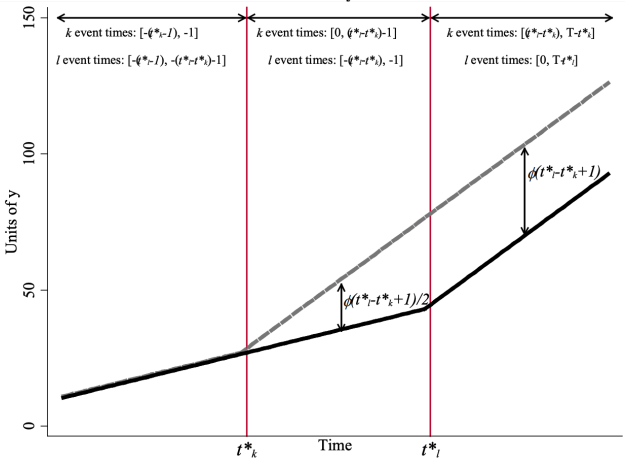
Goodman-Bacon 2x2 comparisons

- The weighting becomes problematic if the effects vary over time – if the effects are instantaneous and time-invariant, the weights are all positive

- However, time-varying effects create bad counterfactual groups, and create negative weights

- Goodman-Bacon provides a way to assess the weights in a given TWFE design

Figure 3. Difference-in-Differences Estimates with Variation in Timing Are Biased When Treatment Effects Vary Over Time



What to do with staggered timing in DinD?

- There's really no reason to use the baseline TWFE in staggered timings
  - A perfect example wherein the estimator does not generate an estimate that maps to a meaningful estimand
- There are different approaches proposed in the literature that are just as good!
  - E.g. de Chaisemartin and d'Haultfoeuille (2020), Callaway and Sant'anna (2020)
- These all are robust to this issue. I find Callaway and Sant'anna quite intuitive, but your circumstances may vary slightly. Key piece to keep in mind that differs a bit:
  - Is my treatment absorbing?
- Irrespective of the exact paper, the key point is that we are generating a counterfactual and need to be careful that our estimator does so correctly

Solutions: Borusyak Hull and Jaravel (2022) Estimator

- Will walk through Sun and Abraham (2021) solution on homework (interacting treatment effects by cohort)

- Callaway and Sant'anna (2021) also provided straightforward solution (not using regression)

- BHJ impute the counterfactual using the not-yet-treated observations

$$Y_{it}(\infty) = \alpha_i + \lambda_t + \epsilon_{it} \tag{8}$$

- Then, we can predict the value for any unit in a time period: $\hat{Y}_{i,t}(\infty)$ and proceed accordingly to construct measures of group by time period ATT ($\mu_{ATT}(g,t) = E(Y_{i,t}(g) - \hat{Y}_{i,t}(\infty)|G_i = g)$

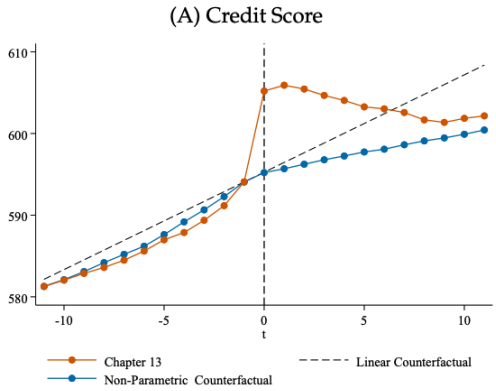
- What is key difference from Callaway and Sant'anna (and why it is more efficient under some settings?)
  - Estimation of α_i uses all the pre-treatment data, rather than just the period before

Aside in event studies

- A key factor in how you construct your counterfactual (and what assumptions you find plausible) are a function of how far into the future you want to estimate outcomes

- An extremely short-run counterfactual could potentially just be a linear extrapolation
  - This assumes that the underlying model is locally linear, rather than globally
  - Construct a counterfactual from just a single time series, but highly non-robust

- Example from a robustness check in my own work (Dobbie et al. 2020)



Constructing a counterfactual is the key goal

- Issue in event study was the attempt to get a "free lunch" – we always need a control group

- Think back to cross-sectional setting with ATT
  - We always knew $Y_i(1)$. Key issue is an estimator for $Y_i(0)$.
  - Event study approaches had issues by ignoring this point and hoping regression would solve problem
  - Notably, this problem disappears if we have full homogeneity + no anticipation and only exclude pre-periods

- Point of emphasis – we need parallel trends to hold to construct a counterfactual in these settings. Why? $Y_{jt}(0) - Y_{j,t-1}(0)$ needs to be a good approximator of $Y_{i,t}(0) - Y_{i,t-1}(0)$.
  - Since we imposed $Y_{it} = \alpha_i + \gamma_t + D_{it}\tau$, the first differencing makes them good approximations

Takeaways

- Staggered timing creates real problems for naive TWFE estimation
- Two distinct issues: negative weights on treatment effects (Goodman-Bacon) and contamination bias in dynamic specifications (Sun and Abraham)
- Solutions exist and are easy to implement: Callaway and Sant'Anna, BHJ imputation, Sun and Abraham interaction-weighted estimator
- The core insight: always think carefully about what counterfactual you are constructing
- Next class: synthetic control, extensions, and a practitioner's checklist